

Ivanna Aleman-Coronado

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EDUCATION

The University of Texas at Austin – Austin, TX

May 2028

B.S. Computer Science, Minor in Robotics

- Relevant Coursework: Computer Architecture, Autonomous Driving, Data Structures, Discrete Math for Computer Science, Calculus II & III,, Robotics.
- Robotics Honors Program, Freshman Research Initiative Participant, Summer Research Fellowship, Hispanics In Computer Science Member

Carroll High School – Southlake, TX

May 2025

PROFESSIONAL EXPERIENCE

Undergraduate Researcher, Autonomous Mobile Robotics Laboratory (AMRL)

Jan 2026 - Present

Austin, Texas

- Conducting research on autonomous vehicle systems under Dr. Nathan Tsoi, focusing on reinforcement learning (PPO), imitation learning, and multi-sensor fusion for scaled autonomous vehicles.
- Developed a Unity simulation environment for coadaptive autonomous driving, integrating merge-tracking metrics and training pipelines via PufferLib on TACC HPC clusters.
- Leveraging a Qwen 2.5-14B LLM to score and evaluate driving behavior, contributing to a multi-agent decision-making framework in dynamic race track environments.

Machine Learning Research Intern, University of Texas at Dallas (StraLiNG Lab)

May 2024 - August 2024

Dallas, Texas

- Developed anomaly-based machine learning models using K-means Clustering, Gaussian Mixture Models, and One-Class classifiers to develop solutions for real-world Denial-of-service attacks.
- Conducted comparative analysis of signature-based versus anomaly-based detection methods on network traffic data, evaluating performance across accuracy, recall, and precision metrics.
- Collaborated with a research team to build a detection system that outperformed traditional signature-based approaches for identifying novel attack patterns.
- Presented findings on machine learning-based network security to UTD faculty.

PROJECTS

Block-Based Programming Game (Squirrel++) | Java, Java Swing | GIMP 3 (level Design)

- Developed an interactive educational game featuring a drag-and-drop visual programming interface that allows players to construct logic-driven programs to control a character in a 2D environment.
- Engineered a custom command system with modular components including movement commands, conditional logic (IF/ELSE statements), and game state conditions, executed through a runtime program interpreter.
- Designed an intuitive UI using Java Swing with layered panels, custom rendering, and dynamically editable conditional blocks that update behavior in real-time.
- Implemented collision detection, state-based movement, and level data evaluation to enable responsive gameplay and real-time decision-making.

Bit Blast | Unity, C#, | Figma (UI Design)

- Developed a 2D falling-block puzzle game that teaches boolean logic (AND, OR, NOT, XOR) through interactive gameplay, deployed to Netlify with a custom domain.
- Engineered a tile-matching system where block behavior is governed by boolean operations, requiring players to reason through logic gate rules to clear the board.
- Designed and implemented all core game systems — block spawning, collision, board evaluation, and win/loss conditions — within a 24-hour build sprint.

AWARDS AND RECOGNITIONS

- **HACS Personal Website Competition winner:** Awarded first place for designing and developing a personal portfolio website, demonstrating technical proficiency and web design.
- **Hispanic Scholarship Fund Scholar:** Selected from 70,000 applications for academic achievement and leadership.
- **National Hispanic Recognition Award:** Honored by the College Board for outstanding achievement among Latino and Hispanic high school students.

TECHNICAL SKILLS

Languages: Java, Python, C, C#, Javascript, Swift, HTML/CSS, React, Javascript

Technologies: Linux, Unity Version Control (Git), VS Code, PyCharm, JFrame/JPanel, FoxGlove, ROS navigation stack.

Work Eligibility: Authorized to work in the U.S.